

Bike paths in Test grid (made-up region)

By Rahul Gurjar, Kogarah

My proposal and its trade-offs

Parking and lanes count four times (heavy) proposal, rank 1

I propose the modelled works in this selection.

I checked the modelled route choices. Useful complete trips and school sites served before, after and newly served are unknown. Entrance links and return trips still need proof. Schools, stations and other places are model goals, not a proved joined network.

Modelled route works are in the selected model totals. Crossing works, existing links retained and remaining route gaps still need checks. I keep these limits beside the lasting space changes.

Modelled parking spaces removed

0

Modelled traffic-lane km removed

0.000

Modelled lower-speed street km

0.000

Parking before, after, added and net change are unknown without an inventory. These loss estimates do not prove the full parking impact.

Capital cost: unknown. Upkeep cost: unknown. No sourced rates or budget are supplied. First delivery stage: unknown; no funded date or build order is proved.

Key gaps: useful trips, school coverage, unique resident gains, usable widths, safe crossing movements and local walking, tree, bus and driveway effects need checks.

I ask council to scope a costed concept design for the selected works and check their usable widths and crossing movements before a build decision. For no new works, I ask council to name a useful route goal and seek its missing public records first. A ranked survey plan tied to those routes is still pending. The owner, approvals and funding are unknown.

Please see the local works and their limits and next decision.

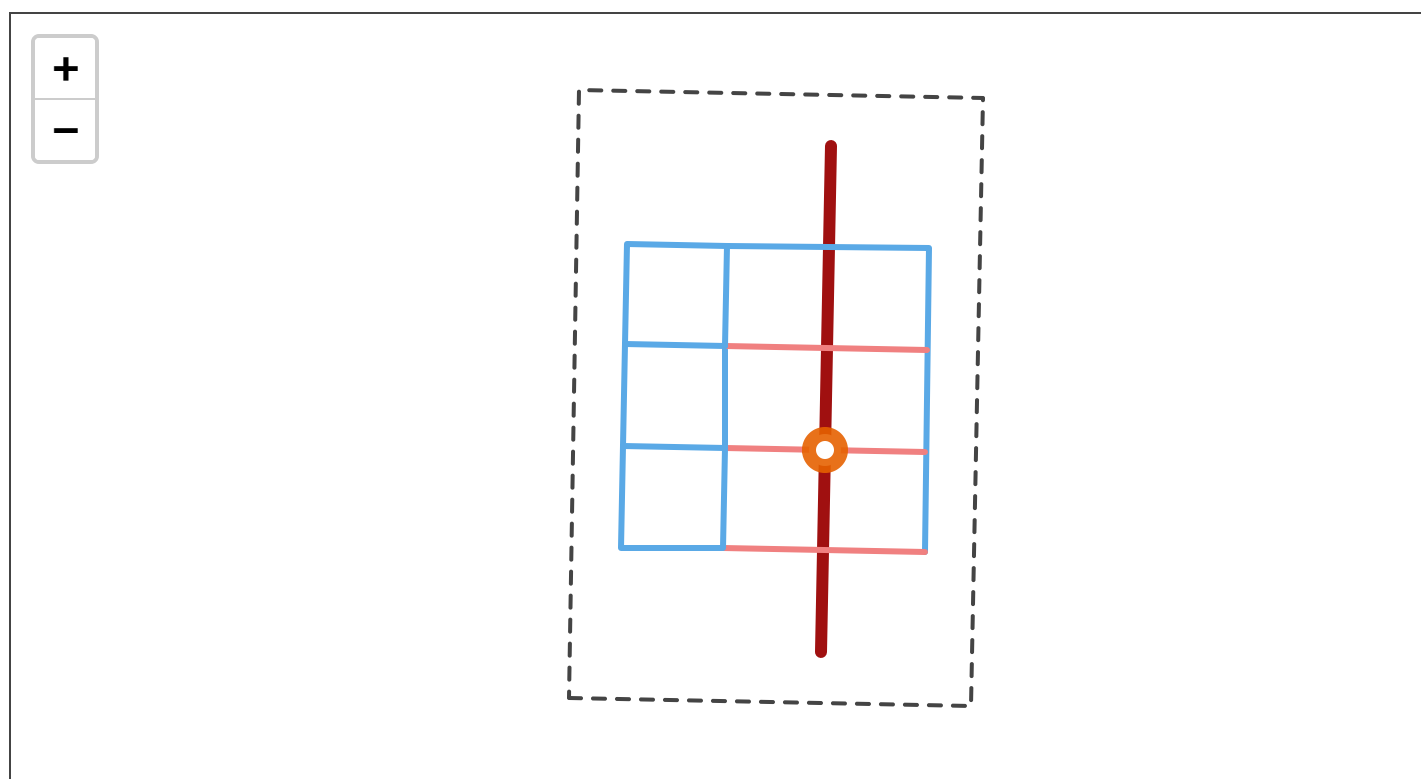
Map of every street a bike may use

Each line is one street segment. The map draws 12 pieces (F4) of street. Hover over a line, or tap it, to read the street name, its type, its stress level and whether it is safe for all ages. The dashed line is the council boundary.

Layers

- ☒ Level one: the calmest (pale blue)
- ☒ Level two: calm (dark blue)
- ☒ Level three: busy (pale red)
- ☒ Level four: the busiest (dark red)
- ☐ Only streets that are safe for all ages
- ☐ Stations (white dots)
- ☐ Schools (yellow dots)
- ☐ Proposed changes
- ☐ Survey options

Survey options need site checks. They are kept out of the confirmed picks.



Options: how much change?

These curves test weights. They may choose the same works; they are not distinct route plans.

The slider opens at my recommended stop for the Weights as set scenario: 1 projects (F13). I stop there because I divide each pick's gain by its cost plus one. I compare that ratio with the best ratio so far, including this pick. I mark the last rank whose ratio is at least 0.25 times that best ratio. I use full precision, before rounding the values shown here. A later pick can meet the rule again. The fixes up to that stop change 0 km (F11) of street. Move the slider to see the cost of doing less or more.

Scenario

- ☐ Parking and lanes count half ☐ Weights as set ☒ Parking and lanes count four times

Totals for the projects picked so far

Access score	<u>100.0</u>
Disruption score	<u>15.0</u>
Parking spaces taken	<u>0</u>
Traffic-lane km taken	<u>0.000</u>
Speed-change km	<u>0.000</u>
People gaining safe reach: aged care	<u>0</u>
People gaining safe reach: college	<u>0</u>
People gaining safe reach: library	<u>0</u>
People gaining safe reach: school	<u>750</u>
People gaining safe reach: station	<u>0</u>
People gaining safe reach: town centre	<u>0</u>
People gaining safe reach: university	<u>0</u>

► Full curve tables and street evidence

Neighbourhood benefits and impacts

Area-level trips and space changes are unknown. Area boundaries and resident unions still need proof. See the [local works](#).

Street plans

Local plans for the selected proposal are pending. The [old minimum-gain shortlist](#) and [its project sheets](#) are separate evidence. They may differ from the selected curve and do not prove its local impacts.

Delivery and next decision

I seek studies and a costed concept design before detailed design or construction. Costs, upkeep, owner, approvals, first build stage and funding are unknown. Dates are not commitments. See my [council ask](#).

Evidence and model limits

I checked [5.2 km \(F1\)](#) of street that a bike may use. [3.0 km \(F2\)](#) meets the model's all-ages criteria. For [0.0 km \(F3\)](#) I have no width. A model result does not prove field safety.

Model scores and type gains follow in the selected totals. Before is the baseline; after is the selected step. Type gains are not a unique resident count. The calm first-leg allowance is a model assumption, not a

confirmed complete trip.

Words I use

Test grid (made-up region)

the council area this report covers.

ADT

average daily traffic: how many motor vehicles use a street in a day.

AAA

All Ages and Abilities: a street that is safe for a child or an older rider.

access

how many needed places a home can reach by bike.

bike

a bicycle, including an electric bicycle.

disruption

what a change takes from people who drive or park today.

Furth

Peter Furth, who wrote the stress tables that I use.

km

a kilometre, which is one thousand metres.

method

the steps and measures used to get a result.

OpenStreetMap

the free map of the world that volunteers keep up to date.

profile

the list of rules and numbers that I use for one country.

project

a set of street and junction changes that I propose together.

reserve

the strip of land beside a street that the public owns.

snapshot

one saved copy of the open data that I read.

council

the local government that runs the streets in an area.

data

facts and numbers that I read from a file.

width

how wide a street or lane is, in metres.

How to check every number

This report uses the data snapshot of 1 October 2026.

F1: Street a bike may use

Value

5.2 km

Method

spec 03. I add up the length of every street segment inside the boundary where a bike may ride.

Files

- segments.csv, sha256
27c0984bcdb2ad112f04c41dfb9bfa5f525a46f380b552cbc0d45187ab2debff

Sources

- OpenStreetMap contributors, licence ODbL 1.0, request: network.osm.gz and boundary.geojson in the snapshot folder

Recipe: run it with `python3 -c` in a folder that holds the release files

```
import csv;print(round(sum(float(r['length_m']) for r in
csv.DictReader(open('segments.csv')) if True)/1000,3))
```

F2: Street that is safe for a child to ride alone (AAA)

Value

3.0 km

Method

spec 04. I score each edge for stress and keep the segments where every edge is AAA.

Files

- segments.csv, sha256
27c0984bcdb2ad112f04c41dfb9bfa5f525a46f380b552cbc0d45187ab2debff

Sources

- OpenStreetMap contributors, licence ODbL 1.0, request: network.osm.gz and boundary.geojson in the snapshot folder

Recipe: run it with `python3 -c` in a folder that holds the release files

```
import csv;print(round(sum(float(r['length_m']) for r in
csv.DictReader(open('segments.csv')) if r['aaa'] == '1')/1000,3))
```

F3: Street with no width estimate

Value

0.0 km

Method

spec 05. I try each width source in rank order. I count the segments where none gave a width.

Files

- segments.csv, sha256
27c0984bcdb2ad112f04c41dfb9bfa5f525a46f380b552cbc0d45187ab2debff

Sources

- OpenStreetMap contributors, licence ODbL 1.0, request: network.osm.gz and boundary.geojson in the snapshot folder

Recipe: run it with python3 -c in a folder that holds the release files

```
import csv;print(round(sum(float(r['length_m']) for r in
csv.DictReader(open('segments.csv')) if r['width_source'] ==
'none')/1000,3))
```

F4: Street piece drawn on the map

Value

12 pieces

Method

spec 13. I join street segments that touch and share a name, street type, stress level, safety and fix. I draw each joined piece once. I count the pieces I drew.

Files

- map.json, sha256
8b1971c195dbb83e05e7dc2ff7dc7f1549ae8c7ad6167eaad1dfc56f67267c85

Sources

- OpenStreetMap contributors, licence ODbL 1.0, request: network.osm.gz and boundary.geojson in the snapshot folder

Recipe: run it with python3 -c in a folder that holds the release files

```
import json;print(len(json.load(open('map.json'))['segments']))
```

F5: Access score at the recommended stop

Value

100.0 points

Method

spec 13. I take the projects in the order I picked them, and add up what each one costs and gains. The shown step is the recommended stop of the Weights as set scenario. The recipe reads that step from frontier.json; change the rank in it to check any other step.

Files

- frontier.json, sha256
0fbef5e8a081b40c3e391760b5b3f15a91062e7f052620ef837681074f50aeca

Sources

- OpenStreetMap contributors, licence ODbL 1.0, request: network.osm.gz and boundary.geojson in the snapshot folder

Recipe: run it with python3 -c in a folder that holds the release files

```
import json;d=json.load(open('frontier.json'));s=[c for c in
d['scenarios'] if c['id']=='shipped'][0];p=s['picks']
[s['recommended_stop'] or 0];print(p['score'])
```

F6: Disruption score at the recommended stop

Value

15.0 points

Method

spec 13. I take the projects in the order I picked them, and add up what each one costs and gains. The shown step is the recommended stop of the Weights as set scenario. The recipe reads that step from frontier.json; change the rank in it to check any other step.

Files

- frontier.json, sha256
0fbef5e8a081b40c3e391760b5b3f15a91062e7f052620ef837681074f50aeca

Sources

- OpenStreetMap contributors, licence ODbL 1.0, request: network.osm.gz and boundary.geojson in the snapshot folder

Recipe: run it with python3 -c in a folder that holds the release files

```
import json;d=json.load(open('frontier.json'));s=[c for c in
d['scenarios'] if c['id']=='shipped'][0];p=s['picks']
[s['recommended_stop'] or 0];print(p['disruption'])
```

F7: Parking spaces taken at the recommended stop

Value

0 spaces

Method

spec 13. I take the projects in the order I picked them, and add up what each one costs and gains. The shown step is the recommended stop of the Weights as set scenario. The recipe reads that step from frontier.json; change the rank in it to check any other step.

Files

- frontier.json, sha256
0fbef5e8a081b40c3e391760b5b3f15a91062e7f052620ef837681074f50aeca

Sources

- OpenStreetMap contributors, licence ODbL 1.0, request: network.osm.gz and boundary.geojson in the snapshot folder

Recipe: run it with python3 -c in a folder that holds the release files

```
import json;d=json.load(open('frontier.json'));s=[c for c in
d['scenarios'] if c['id']=='shipped'][0];p=s['picks']
[s['recommended_stop'] or 0];print(p['parking_spaces'])
```

F8: Traffic-lane length taken at the recommended stop

Value

0.0 km

Method

spec 13. I take the projects in the order I picked them, and add up what each one costs and gains. The shown step is the recommended stop of the Weights as set scenario. The recipe reads that step from frontier.json; change the rank in it to check any other step.

Files

- frontier.json, sha256
0fbef5e8a081b40c3e391760b5b3f15a91062e7f052620ef837681074f50aeca

Sources

- OpenStreetMap contributors, licence ODbL 1.0, request: network.osm.gz and boundary.geojson in the snapshot folder

Recipe: run it with python3 -c in a folder that holds the release files

```
import json;d=json.load(open('frontier.json'));s=[c for c in
d['scenarios'] if c['id']=='shipped'][0];p=s['picks']
[s['recommended_stop'] or 0];print(p['lane_km'])
```

F9: Street length with a lower speed limit at the recommended stop

Value

0.0 km

Method

spec 13. I take the projects in the order I picked them, and add up what each one costs and gains. The shown step is the recommended stop of the Weights as set scenario. The recipe reads that step from frontier.json; change the rank in it to check any other step.

Files

- frontier.json, sha256
0fbef5e8a081b40c3e391760b5b3f15a91062e7f052620ef837681074f50aeca

Sources

- OpenStreetMap contributors, licence ODbL 1.0, request: network.osm.gz and boundary.geojson in the snapshot folder

Recipe: run it with python3 -c in a folder that holds the release files

```
import json;d=json.load(open('frontier.json'));s=[c for c in
d['scenarios'] if c['id']=='shipped'][0];p=s['picks']
[s['recommended_stop'] or 0];print(p['speed_km'])
```

F10: People who gain safe reach at the recommended stop

Value

750.0 people

Method

spec 13. I take the projects in the order I picked them, and add up what each one costs and gains. The shown step is the recommended stop of the Weights as set scenario. The recipe reads that step from frontier.json; change the rank in it to check any other step.

Files

- frontier.json, sha256
0fbef5e8a081b40c3e391760b5b3f15a91062e7f052620ef837681074f50aeca

Sources

- OpenStreetMap contributors, licence ODbL 1.0, request: network.osm.gz and boundary.geojson in the snapshot folder

Recipe: run it with python3 -c in a folder that holds the release files

```
import json;d=json.load(open('frontier.json'));s=[c for c in
d['scenarios'] if c['id']=='shipped'][0];p=s['picks']
[s['recommended_stop'] or 0];print(round(sum(p['people'].values()),3))
```

F11: Street length changed at the recommended stop

Value

0 km

Method

spec 13. I take the projects in the order I picked them, and add up what each one costs and gains. The shown step is the recommended stop of the Weights as set scenario. The recipe reads that step from frontier.json; change the rank in it to check any other step.

Files

- frontier.json, sha256
0fbef5e8a081b40c3e391760b5b3f15a91062e7f052620ef837681074f50aeca

Sources

- OpenStreetMap contributors, licence ODbL 1.0, request: network.osm.gz and boundary.geojson in the snapshot folder

Recipe: run it with python3 -c in a folder that holds the release files

```
import json;d=json.load(open('frontier.json'));s=[c for c in
d['scenarios'] if c['id']=='shipped'][0];p=s['picks']
[s['recommended_stop'] or 0];print(round(sum(p['km_by_fix'].values()),3))
```

F12: Steps drawn on the access against disruption chart

Value

6 steps

Method

spec 13. Each scenario draws one point for no change and one for each project I picked. I count the points of every scenario.

Files

- frontier.json, sha256
0fbef5e8a081b40c3e391760b5b3f15a91062e7f052620ef837681074f50aeca

Sources

- OpenStreetMap contributors, licence ODbL 1.0, request: network.osm.gz and boundary.geojson in the snapshot folder

Recipe: run it with python3 -c in a folder that holds the release files

```
import json;print(sum(len(c['picks']) for c in
json.load(open('frontier.json'))['scenarios']))
```

F13: Projects up to the recommended stop

Value

1 projects

Method

spec 13. I divide each pick's gain by its cost plus one, using full precision. I compare it with the best ratio so far, including this pick. The stop is the last rank at least 0.25 times that best ratio, even after a weaker pick.

Files

- frontier.json, sha256
0fbef5e8a081b40c3e391760b5b3f15a91062e7f052620ef837681074f50aeca

Sources

- OpenStreetMap contributors, licence ODbL 1.0, request: network.osm.gz and boundary.geojson in the snapshot folder

Recipe: run it with python3 -c in a folder that holds the release files

```
import json;d=json.load(open('frontier.json'));s=[c for c in
d['scenarios'] if c['id']=='shipped'][0];p=s['picks']
[s['recommended_stop'] or 0];print(s['recommended_stop'] or 0)
```

Summary

Access score 25.0 before and 100.0 after 1 projects.

Projects picked, by kind

Kind	Projects
corridor	0
neighbourhood	0
route	1

Corridor candidates

Outcome	Candidates
made	0
picked	0
not picked	0

Kilometres by fix

Fix	km
none	0.000

Disruption

Item	Total
Parking spaces removed	0
Car lane km taken	0.0
Speed limit km lowered	0.0
Signals added	1
Refuges added	0

People who gain safe reach, by place type

Place type	People
aged_care	0
college	0
library	0
school	750
station	0
town_centre	0
university	0

Projects, ranked

Rank	Project	Gain	Score after	People gaining safe reach
1	Grid School: unnamed residential	75.0	100.0	750

Project sheets

Grid School: unnamed residential

Gain 75.0. Score after 100.0.



The same map as a table

Street				Fix		
unnamed residential				signals		
Element	Street	Fix	Model margin	Width source	Width confidence / Source confidence	Length m
junction: 1010	unnamed residential	signals	holds up	lanes	low	0.0

Disruption totals

Item	Total
Parking spaces removed	0
Car lane km taken	0.0
Speed limit km lowered	0.0
Signals added	1
Refuges added	0

Kilometres by fix

Fix	km
none	0.0

Street cross-sections

unnamed residential; width source lanes, source confidence low

Before cross-section



The same strips as a table

parking: 2.1 m
through: 3.0 m
through: 3.0 m
parking: 2.1 m
spare: 0.6 m

After cross-section



The same strips as a table

parking: 2.1 m
through: 3.0 m
through: 3.0 m
parking: 2.1 m
spare: 0.6 m

unnamed residential; width source lanes, source confidence low

Before cross-section



The same strips as a table

parking: 2.1 m
through: 3.0 m
through: 3.0 m
parking: 2.1 m
spare: 0.6 m

After cross-section



The same strips as a table

parking: 2.1 m
through: 3.0 m
through: 3.0 m
parking: 2.1 m
spare: 0.6 m

Check location: [Street View](#) · [Mapillary](#)

Place type	People
aged_care	0
college	0
library	0
school	750
station	0
town_centre	0
university	0

Method

1. The tool reads one snapshot of OpenStreetMap and the other open data named under credits. It makes no network call.
2. It builds the street network and gives every street a stress level from 1 to 4 by the Furth tables. A street is safe for all ages when it meets the AAA rules of the profile.
3. It estimates the width of each street from lane tags, width tags and the land reserve, and keeps the best source. It then tests six fixes against that width.
4. It measures access: for each home it checks which places a person can reach on safe streets within the reach limit, and scores that by place type.
5. It picks projects one at a time. Each project is the set of street and junction fixes with the best gain in access for its cost, until no project gains enough.

Profile values and assumptions

Rows marked assumption have no published source.

Value	Setting	Source	Kind
aaa.mixed_traffic[0].max_speed_kmh	30	TfNSW Cycleway Design Toolbox (2020)	sourced
aaa.mixed_traffic[0].max_adt	2000	TfNSW Cycleway Design Toolbox (2020)	sourced
aaa.painted_lanes_count	False	TfNSW Cycleway Design Toolbox (2020): painted lanes are not separation	sourced
widths_m.one_way_cycleway.min	1.5	Austroads Guide to Road Design Part 6A	sourced
widths_m.one_way_cycleway.desirable	2.0	TfNSW Cycleway Design Toolbox (2020)	sourced
widths_m.two_way_cycleway.min	2.5	Austroads Guide to Road Design Part 6A	sourced
widths_m.two_way_cycleway.desirable	3.0	TfNSW Cycleway Design Toolbox (2020)	sourced
widths_m.separator_traffic.min	0.5	TfNSW Cycleway Design Toolbox (2020)	sourced
widths_m.separator_traffic.desirable	1.0	TfNSW Cycleway Design Toolbox (2020)	sourced
widths_m.separator_parking.min	1.0	TfNSW Cycleway Design Toolbox (2020)	sourced
widths_m.shared_path.min	3.0	Austroads Guide to Road Design Part 6A	sourced
widths_m.shared_path.desirable	4.0	TfNSW Cycleway Design Toolbox (2020)	sourced

Value	Setting	Source	Kind
widths_m.traffic_lane.min	3.0	Austroads Guide to Road Design Part 3 (60 km/h or less)	sourced
widths_m.parking_lane	2.1	AS 2890.5	sourced
widths_m.verge_default	3.5	assumption: typical Sydney footpath plus nature strip	assumption
parking.bay_length_m	6.0	assumption: AS 2890.5 parallel bay plus gaps	assumption
parking.driveway_share	0.3	assumption: share of kerb lost to driveways on a local street	assumption
road_diet.max_adt	20000	FHWA Road Diet Informational Guide (2014)	sourced
quietway.target_speed_kmh	30	TfNSW Cycleway Design Toolbox (2020)	sourced
fit.prefer_separation	True	assumption: a council can build a separated lane on its own local roads, but a 30 km/h limit needs Transport for NSW approval (NSW Speed Zoning Guidelines)	assumption
fit.speed_approval_body	Transport for NSW	NSW Speed Zoning Guidelines	sourced
crossing.refuge_min_m	1.8	Mekuria, Furth and Nixon (2012), Table 8 (six feet)	sourced

Value	Setting	Source	Kind
grade.steep_pct	5	AASHTO Guide for the Development of Bicycle Facilities (2012), grades steeper than 5 percent are hard to ride	sourced
grade.min_length_m	100	assumption: a climb under 100 m is over before it tires most riders	assumption
road_classes.living_street.speed_kmh	10	assumption: typical posted limit for the class	assumption
road_classes.living_street.adt	200	assumption: typical daily traffic for the class	assumption
road_classes.living_street.lanes	1	assumption: typical lane count for the class when untagged	assumption
road_classes.living_street.parking	True	assumption: kerb parking is usual on this class when parking is untagged	assumption
road_classes.primary.speed_kmh	60	assumption: typical posted limit for the class	assumption
road_classes.primary.adt	25000	assumption: typical daily traffic for the class	assumption
road_classes.primary.lanes	4	assumption: typical lane count	assumption

Value	Setting	Source	Kind
		for the class when untagged	
road_classes.primary.parking	False	assumption: kerb parking is not usual on this class when parking is untagged	assumption
road_classes.residential.speed_kmh	50	assumption: NSW Road Rules default urban limit of 50 km/h	assumption
road_classes.residential.adt	750	assumption: typical daily traffic for the class	assumption
road_classes.residential.lanes	2	assumption: typical lane count for the class when untagged	assumption
road_classes.residential.parking	True	assumption: kerb parking is usual on this class when parking is untagged	assumption
road_classes.secondary.speed_kmh	60	assumption: typical posted limit for the class	assumption
road_classes.secondary.adt	12000	assumption: typical daily traffic for the class	assumption
road_classes.secondary.lanes	2	assumption: typical lane count for the class when untagged	assumption
road_classes.secondary.parking	True	assumption: kerb parking is usual on this class when	assumption

Value	Setting	Source	Kind
		parking is untagged	
road_classes.service.speed_kmh	20	assumption: typical posted limit for the class	assumption
road_classes.service.adt	200	assumption: typical daily traffic for the class	assumption
road_classes.service.lanes	1	assumption: typical lane count for the class when untagged	assumption
road_classes.service.parking	False	assumption: kerb parking is not usual on this class when parking is untagged	assumption
road_classes.tertiary.speed_kmh	50	assumption: NSW Road Rules default urban limit of 50 km/h	assumption
road_classes.tertiary.adt	5000	assumption: typical daily traffic for the class	assumption
road_classes.tertiary.lanes	2	assumption: typical lane count for the class when untagged	assumption
road_classes.tertiary.parking	True	assumption: kerb parking is usual on this class when parking is untagged	assumption
road_classes.trunk.speed_kmh	70	assumption: typical posted	assumption

Value	Setting	Source	Kind
		limit for the class	
road_classes.trunk.adt	35000	assumption: typical daily traffic for the class	assumption
road_classes.trunk.lanes	4	assumption: typical lane count for the class when untagged	assumption
road_classes.trunk.parking	False	assumption: kerb parking is not usual on this class when parking is untagged	assumption
road_classes.unclassified.speed_kmh	50	assumption: NSW Road Rules default urban limit of 50 km/h	assumption
road_classes.unclassified.adt	1500	assumption: typical daily traffic for the class	assumption
road_classes.unclassified.lanes	2	assumption: typical lane count for the class when untagged	assumption
road_classes.unclassified.parking	True	assumption: kerb parking is usual on this class when parking is untagged	assumption
implicit_speeds.AU:urban	50	OpenStreetMap Wiki, Default speed limits: AU:urban	sourced
implicit_speeds.AU:rural	100	OpenStreetMap Wiki, Default	sourced

Value	Setting	Source	Kind
		speed limits: AU:rural	
implicit_speeds.AU: motorway	110	OpenStreetMap Wiki, Default speed limits: AU:motorway	sourced
implicit_speeds.GB: nsl_restricted	48.28	OpenStreetMap Wiki, Default speed limits: GB:nsl_restricted	sourced
implicit_speeds.GB: nsl_single	96.56	OpenStreetMap Wiki, Default speed limits: GB:nsl_single	sourced
implicit_speeds.GB: nsl_dual	112.65	OpenStreetMap Wiki, Default speed limits: GB:nsl_dual	sourced
implicit_speeds.GB: motorway	112.65	OpenStreetMap Wiki, Default speed limits: GB:motorway	sourced

Data credits

- Made-up test data from scripts/make_test_grid.py, licence CC0 1.0: None needed, CC0 1.0
- OpenStreetMap contributors, licence ODbL 1.0: © OpenStreetMap contributors, ODbL 1.0

Rebuild this report

Snapshot 2026-10-01. Config hash

1258a6fbf083fcd15143bc4dad094757183c46e070315db276920bdbaa24511b.

```
uv run bikeplan snapshot pull snapshots/test-grid/2026-10-01/manifest.json
uv run bikeplan run regions/test-grid.yaml --snapshot data/cache/test-grid/2026-10-01 --out out
uv run bikeplan verify out
```